PS-019 Enterprise Knowledge Retention & Discovery Assistant

PS-019

Enterprise Knowledge Retention & Discovery Assistant

ION India Hackathon 2026 - CAT & Commodities Edition | 19-20 June 2026

Submitted by	Sushil Rakate
Location	Pune
Product / System	All / enterprise knowledge sources
Business Impact	High
Technical Uncertainty	Medium
Type	Internal Productivity
PoC Feasible in 24h	Yes - a working prototype is achievable

Problem Statement

When people move on, organizational intelligence stays, learns, and grows.

Organizations spend years building expertise through employees, customer engagements, architectural decisions, incidents, projects, support interactions, and operational experiences.

Unfortunately, much of this knowledge remains scattered across numerous systems and often resides only in the minds of experienced employees.

The opportunity is to build a governed knowledge discovery assistant that can search, connect, and summarize enterprise knowledge across approved sources while respecting access control and giving users reusable, source-backed answers.

> **Current workaround:** *Teams rely on scattered documentation, knowledge bases, chat histories, emails, recorded meetings, subject matter experts. And are you only able to discover where the document is?*
>
> *Employees need answers to:*

> - *Why was this decision made?*
> - *How have we solved this before?*
> - *Who are the experts in this area?*
> - *What lessons did previous teams learn?*
> - *What risks should we consider?*
> - *Customer insights?*

> *Without this context:*
>
> > - *Decision-making quality decreases.*
> > - *Knowledge is repeatedly recreated.*
> > - *Organizations become dependent on a few key individuals.*

As a result, organizations repeatedly invest time and money in rediscovering knowledge that already existed.

The Organizational Brain aims to solve this challenge by creating a continuously evolving AI-powered intelligence layer that captures, connects, preserves, and surfaces organizational expertise across people, systems, and business functions.

Rather than simply finding documents, employees can interact with the organization's accumulated intelligence through natural language and receive contextual answers, recommendations, decision history, expert guidance, and lessons learned.

The Opportunity

Create an AI-powered Organizational Brain capable of:

Preserving Expertise

Capture critical knowledge before it disappears.

Connecting Information

Link information across systems, teams, projects, products, and decisions.

Building Organizational Memory

Retain historical context and reasoning.

Enabling Decision Intelligence

Help employees understand:

- What happened
- Why it happened
- What alternatives existed
- What outcomes occurred

Creating Digital Experts

Simulate institutional expertise based on historical organizational behavior.

***Scope guidance:* Focus on knowledge retention and reuse: find relevant prior knowledge, summarize it clearly, show provenance, and highlight where the answer is incomplete or needs SME validation.**

***The hackathon should focus on proving the core concept rather than building the complete enterprise platform.* Primary Goal should be to demonstrate that enterprise knowledge can be:**

- ***Captured***
- ***Connected***
- ***Queried***
- ***Reasoned over***
- ***Presented contextually***

Using AI over 3-4 data sources like SharePoint documents, Team meeting transcripts, JIRA tickets, Confluence pages.

Measurable Impact

Metric	*Target / Figure*	*Interpretation*
Information search time	***Reduced time spent searching across systems***	***Employees should find relevant knowledge faster instead of manually checking multiple repositories.***
Duplicate effort	***Reduced repeated analysis and investigation***	***Teams should avoid re-solving problems where prior decisions, incidents, or discussions already exist.***
Onboarding speed	***Shorter onboarding cycles***	***New joiners should access contextual knowledge without depending only on SMEs.***
SME dependency	***Lower dependency on key individuals***	***Knowledge remains reusable even when experienced employees are unavailable or leave.***
Decision velocity	***Faster decisions***	***Teams can reuse documented rationale, past outcomes, and incident learnings.***

Value to ION

For ION, this capability can preserve institutional memory, reduce productivity loss from knowledge attrition, and improve day-to-day execution across teams. It turns scattered enterprise knowledge into a reusable asset rather than something dependent on individuals.

This Organizational Brain ensures that every lesson learned, decision made, problem solved, and insight discovered becomes a permanent asset that strengthens the organization rather than being lost over time.

Value Area	Expected Benefit
Knowledge retention	Capture and reuse important context even when employees move roles or leave.
Internal productivity	Reduce manual searching across documentation, chats, tickets, emails, and meeting notes.
Onboarding	Help new team members understand historical decisions, known issues, workflows, and ownership faster.
Operational consistency	Reduce variation in answers by grounding responses in approved internal sources.
Enterprise scalability	Create a pattern that can be expanded across products, teams, and knowledge of repositories.

Target Users

The primary users are team members across products and functions who need to find and reuse organizational knowledge. This includes new joiners, support teams, engineering teams, product teams, implementation teams, operations teams, and managers who need fast access to prior decisions or known solutions.

The solution also benefits SMEs by reducing repeated questions and allowing their knowledge to be reused through curated documents, discussions, tickets, and transcripts.

Scope for This Hackathon

The recommended 24-hour PoC should focus on a constrained but realistic knowledge domain, such as SharePoint documents, Teams meeting transcripts, Jira tickets, Confluence pages. The PoC should demonstrate retrieval, summarization, provenance, and access-aware output using mock or approved data.

Level	*Description*	Examples
Level 0	*Manual knowledge discovery*	Employees manually search SharePoint, Confluence, Jira, Teams, emails, meeting recordings, and ask SMEs. Current workaround.
Level 1	*Single-source Q&A*	The assistant answers questions from one curated repository or document set. Baseline target.
Level 2	*Multi-source knowledge retrieval*	The assistant retrieves relevant context from multiple mock or approved sources and summarizes with references. Strong PoC target.
Level 3	*Access-aware knowledge assistant*	Assistant respects user context, source permissions, freshness, and confidence. Stretch scope.
Level 4	*Enterprise knowledge memory platform*	Full-scale enterprise indexing, governance, source connectors, retention policy, and production access controls. Out of scope for 24 hours.

Recommended Scope

- Select one constrained knowledge domain and create a small representative corpus.
- Support natural-language questions from employees.
- Retrieve relevant snippets/documents from two or more mock or approved knowledge sources where feasible.
- Generate a concise answer with source references, confidence, and assumptions.
- Highlight missing or conflicting knowledge and suggest the next source or SME to consult.
- Show how access control would be respected, even if implemented as mocked roles/permissions for the PoC.

Out of Scope

- Indexing every enterprise repository in 24 hours.
- Using confidential, restricted, or personal data without explicit approval.
- Bypassing source-system permissions or access controls.
- Replacing official documentation owners or SMEs for high-risk decisions.
- Production-grade retention governance, legal hold, compliance, or records-management implementation.

How the Solution Improves on Existing Workflows

- **Unified discovery:** Gives employees one guided entry point instead of manually searching multiple tools.
- **Knowledge reuse:** Surfaces prior answers, incident learnings, decisions, and documents that would otherwise remain hidden.
- **Context preservation:** Keeps rationale, links, ownership, and related artefacts attached to the answer.
- **SME load reduction:** Deflects repeated questions by making existing SME knowledge discoverable.
- **Onboarding acceleration:** Helps new team members understand product, team, and process context faster.
- **Trust and governance:** Shows sources, confidence, and access assumptions instead of producing unsupported answers.

Supporting Materials / References

The following inputs may be used where available and approved for participant access:

- SharePoint pages or exported documents.
- Confluence pages or mock knowledge base articles.
- Jira tickets, issue summaries, decision records, or incident reports.
- Teams conversation exports, meeting transcripts, or summarized discussion notes where approved.
- AI chat histories or prior Q&A examples were approved.
- Source code repository README files, architecture notes, or troubleshooting docs.
- GBrain project context as a reference project

Recommended PoC Approach

Build an Enterprise Knowledge Retention & Discovery Assistant that answers employee questions using a small approved/mock enterprise knowledge corpus. The PoC should show retrieval, summarization, source traceability, and a basic permission model.

Component	Purpose
Input layer	Employee question, selected role/team, optional product or topic filter.
Connector/mock source layer	Mock or approved SharePoint, Confluence, Jira, Teams, transcript, document, or repository content.
Index/retrieval layer	Search or embedding-based retrieval over the selected corpus.
Answer generation layer	Summarize relevant knowledge into a concise, reusable answer.
Provenance layer	Display source references, dates, owners, confidence, and gaps.
Access-control layer	Filter or label results based on mocked user role/source permissions.
Feedback layer	Allow the user to mark answer quality, missing sources, or SME validation needed.

Example Plan

1. Choose a focused domain such as onboarding, incident learnings, or product troubleshooting.
2. Create 20-50 mock or approved knowledge records across 2-3 source types.
3. Build a simple UI where the user asks a natural-language question and selects a role/team.
4. Retrieve the most relevant records and show snippets with source links or IDs.
5. Generate a concise answer with references, assumptions, confidence, and missing information.
6. Demonstrate role-based filtering by showing that restricted sources are hidden or flagged.
7. Show feedback capture for “helpful/not helpful”, missing source, or SME validation required.
8. Explain how the approach would scale to additional repositories and governance controls after the hackathon.

Suggested APIs / Interfaces

API / Interface	Purpose
POST /knowledge/query	Accepts questions, user context, product/topic filters, and returns a grounded answer.
GET /knowledge/sources	Lists available to mock or approved knowledge sources for the PoC.
POST /knowledge/index	Indexes approved documents or mock records for retrieval.
GET /knowledge/result/{id}	Returns answer details, source references, confidence, and gaps.
POST /knowledge/feedback	Captures answer feedback, missing sources, and SME validation requests.
GET /knowledge/access-check	Simulates or checks whether the user can view a given source/result.

Constraints and Considerations

- **Data privacy:** Use mock, sanitized, or explicitly approved data only. Do not expose confidential content, personal data, secrets, credentials, or restricted customer information.
- **Access control:** The assistant must not reveal content the user is not entitled to view. For the PoC, roles and permissions may be mocked but should be visible.
- **Source trust:** Answers should show source references and distinguish verified content from assumptions or stale information.
- **Knowledge freshness:** The PoC should show source dates or freshness indicators where available.
- **Hallucination risk:** The assistant should say when it does not know and point to missing sources or SME validation needs.
- **Enterprise scale:** Full production-scale indexing across all systems is not expected in 24 hours.

What Success Looks Like

A strong submission will demonstrate a working assistant that helps employees find and reuse organizational knowledge from a focused corpus, while making source traceability and access assumptions clear.

At minimum, the PoC should show

- A small but realistic knowledge corpus from at least one approved or mock enterprise source.
- A natural-language query experience for employees.
- Relevant retrieval and summarized answer generation.
- Visible source references or links/IDs for every answer.
- A basic access-control or role-filtering demonstration.
- A way to flag missing, stale, or conflicting knowledge.
- A clear explanation of how the solution reduces search time, duplicate effort, onboarding delays, and SME dependency.

Beyond the PoC, the team should be able to answer

- Which repositories should be connected first and why?
- How will source permissions be respected end-to-end?
- How will stale or conflicting knowledge be detected and owned?
- How will the answer quality and productivity gains be measured?
- What is the backlog item that comes next?

The definition of done is not a full enterprise memory platform. It is a focused, demonstrable MVP that proves knowledge of retrieval, source-backed summarization, access awareness, and reuse value.

Future Scope and Scalability

Phase	Functionality	Capability
1	Curated knowledge Q&A	Answer questions from a small, approved corpus with source references.
2	Multi-source retrieval	Connect multiple repositories and unify search across documents, tickets, chats, and transcripts.
3	Access-aware enterprise assistant	Apply role-based access, entitlement checks, source freshness, and audit logging.
4	Knowledge lifecycle management	Detect stale/conflicting content, suggest owners, and route updates or SME validation.
5	Enterprise knowledge memory platform	Reusable organizational memory layer across products, teams, and workflows.

Longer-term opportunities include source connectors, permission-aware retrieval, freshness scoring, duplicate knowledge detection, SME validation workflows, onboarding packs, incident-learning libraries, and integration with enterprise copilots or GBrain-style initiatives.

For any questions on this problem statement, write to: core-ionindiahackathon2026@iongroup.com